Chapter 23 · Semester 3

GEOMETRICAL OPTICS

Notes by Cikgu Suhaili

33 worked examples 4 simulations Exercises with answers

Geometrical optics: introduction

Geometrical optics studies light on the assumption that it travels in straight lines called **rays**. A collection of rays is a **beam**. Two laws describe how light changes direction at a surface: the law of reflection and the law of refraction.

Real image	Virtual image
formed by light rays that actually pass through it	the rays only appear to come from it; no light passes through it
can be formed on a screen	cannot be formed on a screen
drawn with solid lines	drawn with dashed lines

> **Law of reflection**
>
> the angle of reflection equals the angle of incidence, and the incident ray, the reflected ray and the normal lie in the same plane. (Angles are measured from the normal, not from the surface.)

A **plane mirror** forms an image that is virtual, upright, the same size ($m = +1$), as far behind the mirror as the object is in front ($v = u$), and laterally inverted. This is why “AMBULANCE” is written back to front on the front of an ambulance: it reads correctly in a driver’s mirror.

23.1 Spherical mirrors

A spherical mirror is a reflecting surface that is part of a sphere. It is **concave** if the reflection is on the inner surface and **convex** if it is on the outer surface.



Figure 23.1: C = centre of curvature, P = pole (vertex), F = focal point, CP = principal axis, AB = aperture, f = focal length (FP), r = radius of curvature (CP).

Focal point of a concave mirror

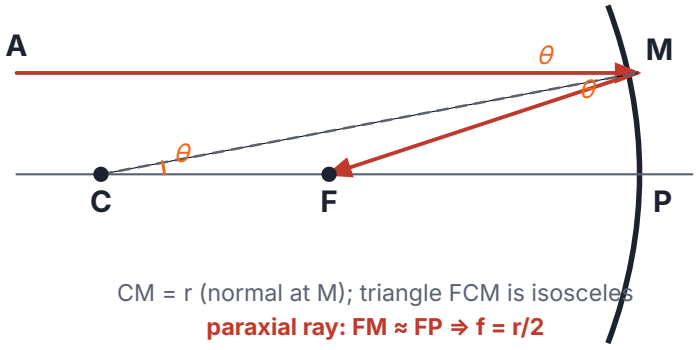
the point on the principal axis **through which rays parallel and close to the principal axis pass after reflection**.

Focal point of a convex mirror

the point on the principal axis **from which rays parallel and close to the principal axis appear to diverge after reflection**.

This is true only for **paraxial rays** (rays close to the principal axis). Wide rays are focused nearer the mirror (spherical aberration).

Relation between f and r



CM = r (normal at M); triangle FCM is isosceles

paraxial ray: FM ≈ FP ⇒ f = r/2

Figure 23.2: a paraxial ray AM, parallel to the axis, reflects through F.

CM is the normal at M, so the angle of incidence = angle of reflection = θ.

AM ∥ CP, so ∠MCF = θ (alternate angles). Triangle FCM has two equal angles, so it is isosceles: FC = FM.

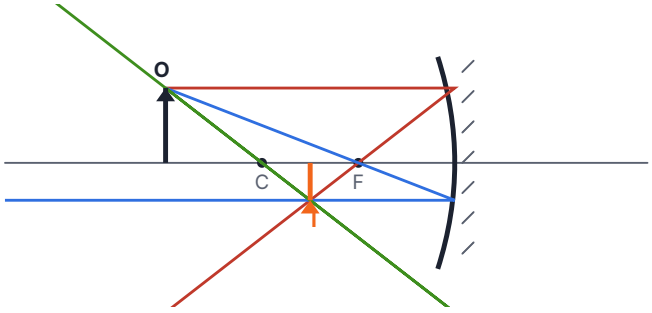
For a paraxial ray, M is very close to P, so FM ≈ FP. Then FC = FP, i.e. F is the midpoint of CP.

$$f = \frac{r}{2}$$

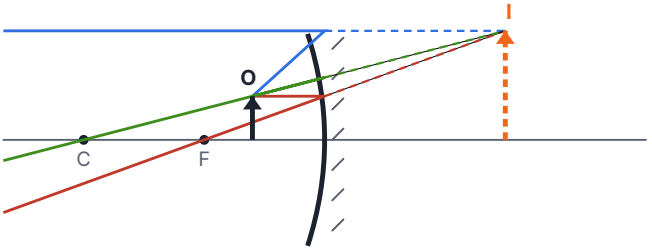
Ray diagrams

- **Ray 1**: a ray parallel to the principal axis is reflected through F (concave), or as if it came from F (convex).
- **Ray 2**: a ray through F (or heading towards F, convex) is reflected parallel to the axis.
- **Ray 3**: a ray through C (or heading towards C, convex) is reflected back along the same path.
- Any two rays from the top of the object locate the top of the image.

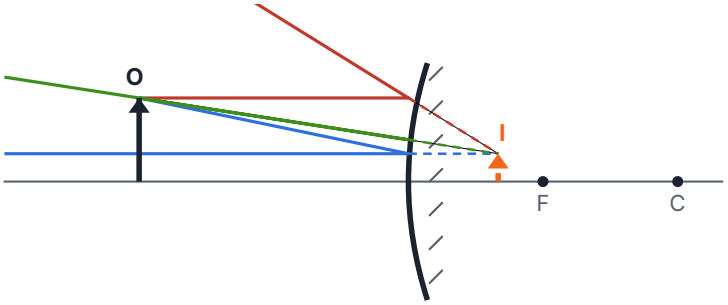
Object position (concave mirror)	Image position	Nature
at infinity	at F	real, inverted, diminished (a point)
beyond C	between C and F	real, inverted, diminished
at C	at C	real, inverted, same size
between C and F	beyond C	real, inverted, magnified
at F	at infinity	—
between F and P	behind the mirror	virtual, upright, magnified



Concave mirror, object beyond C.



Concave mirror, object between F and P: virtual, upright, magnified.



Convex mirror: the image is always virtual, upright and diminished, between P and F behind the mirror.

The mirror equation

Using similar triangles in the ray diagram (object distance OP = u, image distance IP = v, CP = r):

Ray to P: $\frac{h}{u} = \frac{h'}{v} \Rightarrow \frac{h'}{h} = \frac{v}{u}$ ... (1)

Ray through C: $\frac{h}{u-r} = \frac{h'}{r-v} \Rightarrow \frac{h'}{h} = \frac{r-v}{u-r}$ ... (2)

(1) = (2): $v(u-r) = u(r-v) \Rightarrow 2uv = r(u+v) \Rightarrow \frac{2}{r} = \frac{1}{u} + \frac{1}{v}$

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f} = \frac{2}{r} \qquad m = \frac{h'}{h} = -\frac{v}{u}$$

m = linear magnification

Quantity	Positive (+)	Negative (−)
object distance u	real object	virtual object
image distance v	real image (in front of the mirror)	virtual image (behind the mirror)
focal length f, radius r	concave mirror	convex mirror
magnification m	upright image	inverted image

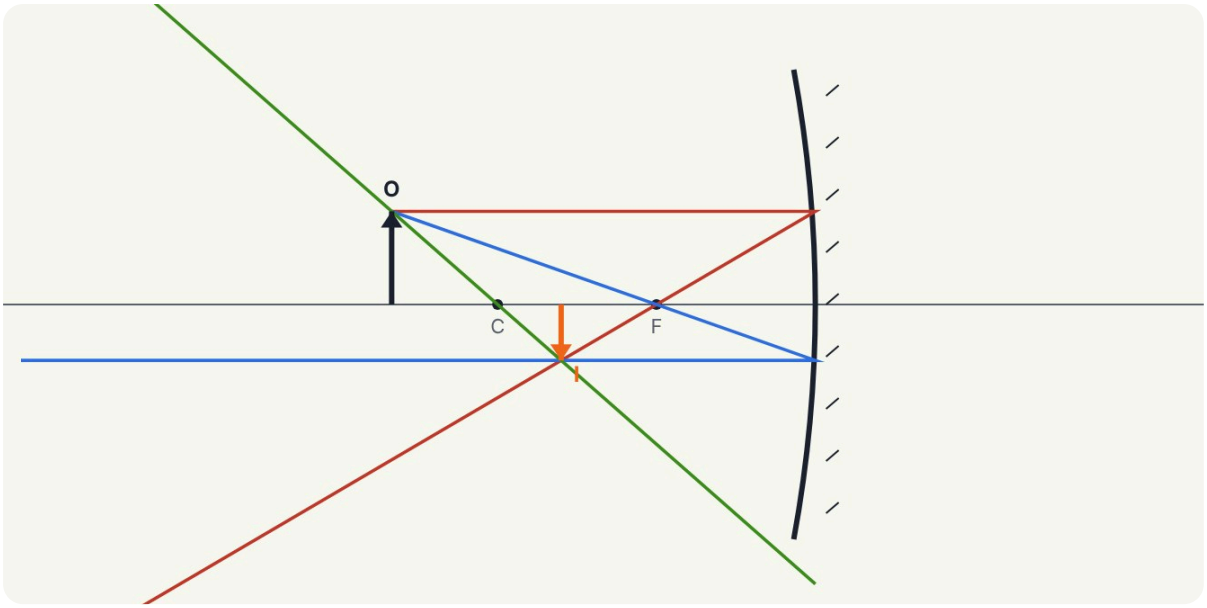
Teacher's note

Write the sign of f first (concave +, convex −), solve for v, then describe the image from the signs: $v > 0$ real, $v < 0$ virtual; $m > 0$ upright, $m < 0$ inverted; $|m| > 1$ magnified.

Simulation

Ray diagram for a spherical mirror

Choose a concave or convex mirror and drag the object along the axis. The three principal rays are drawn and the image appears where they meet. Watch the image change from real and inverted to virtual and upright as the object passes F.



f	image distance v	magnification m	image
15.0 cm	**24.0 cm**	**−0.60**	**real, inverted, diminished**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 23 to use the simulation.

Example 23.1.1

An object 6 cm high is 30 cm in front of a convex mirror of radius 40 cm. Find the position and height of its image.

Solution

Convex: $r = -40$ cm; $\frac{2}{r} = \frac{1}{u} + \frac{1}{v} \Rightarrow \frac{2}{-40} = \frac{1}{30} + \frac{1}{v} \Rightarrow v =$ **−12 cm** (virtual, 12 cm behind the mirror)

$m = -v/u = 12/30 = 0.40 \Rightarrow h' = 0.40 \times 6 =$ **2.4 cm** (upright)

Example 23.1.2

An object is 15 cm from (a) a concave mirror, (b) a convex mirror, each of radius of curvature 20 cm. Find the image position and magnification in each case.

Solution

(a) $f = +10$ cm: $\frac{1}{v} = \frac{1}{10} - \frac{1}{15} \Rightarrow v =$ **+30 cm** (real, 30 cm in front); $m = -30/15 =$ **−2** (inverted, twice as high)

(b) $f = -10$ cm: $\frac{1}{v} = \frac{1}{-10} - \frac{1}{15} \Rightarrow v =$ **−6.0 cm** (virtual, 6.0 cm behind); $m = 6/15 =$ **+0.40** (upright, two-fifths as high)

Example 23.1.3

What is the focal length of a convex mirror that forms an image one-sixth the size of an object 12 cm from the mirror?

Solution

Convex mirror: image virtual and upright, $m = +1/6 \Rightarrow v = -u/6 = -2$ cm

$\frac{1}{f} = \frac{1}{12} + \frac{1}{-2} \Rightarrow f =$ **−2.4 cm**

Example 23.1.4

An object 20 cm from a concave mirror forms a real image three times its size. (a) Find the focal length. (b) Where must the object be placed to give a virtual image three times its size?

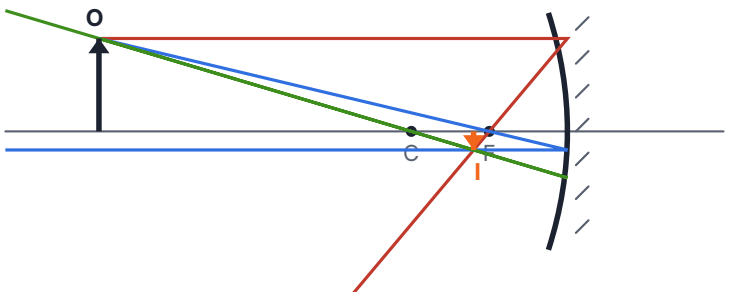
Solution

(a) real ⇒ inverted, $m = -3 \Rightarrow v = 3u = 60$ cm; $\frac{1}{f} = \frac{1}{20} + \frac{1}{60} \Rightarrow f =$ **+15 cm**

(b) virtual ⇒ upright, $m = +3 \Rightarrow v = -3u$; $\frac{1}{u} - \frac{1}{3u} = \frac{1}{15} \Rightarrow u =$ **10 cm**

Example 23.1.5

An object 2.0 cm high is 30 cm from a concave mirror of radius of curvature 10 cm. Find the position and the characteristics of the image.



Solution

$f = r/2 = +5$ cm; $\frac{1}{v} = \frac{1}{5} - \frac{1}{30} \Rightarrow v =$ **+6.0 cm**

$m = -6/30 = -0.20$, $h' = -0.40$ cm

The image is **real**, **inverted**, **diminished** (0.4 cm high), 6.0 cm in front of the mirror.

Exercises: spherical mirrors

1. A concave mirror has a focal length of 10 cm. Find the two positions where an object can be placed to give an image twice the height of the object.

   Answer

   15 cm (real, inverted image) and 5.0 cm (virtual, upright image)

2. A convex mirror of radius of curvature 40 cm forms an image half the height of the object. Find the object and image positions.

   Answer

   u = 20 cm; image 10 cm behind the mirror (v = −10 cm)

3. An object is 5.0 cm in front of a concave mirror of focal length 10.0 cm. Find the location of the image and its characteristics.

   Answer

   v = −10 cm (10 cm behind the mirror); m = +2: virtual, upright, twice the height

4. What kind of spherical mirror, and what radius, gives an upright image one-fifth as large as an object placed 15 cm in front of it?

   Answer

   m = +1/5 ⇒ v = −3.0 cm, f = −3.75 cm: a convex mirror of radius 7.5 cm (r = −7.5 cm)

5. A concave mirror forms an image, on a wall 3.0 m from the mirror, of a lamp filament 10 cm in front of the mirror. (a) What is the radius of curvature? (b) How high is the image if the filament is 5 mm high?

   Answer

   (a) f = 9.68 cm, r = 19.4 cm (b) m = −30, image 15 cm high (inverted)

6. What are the nature, size and location of the image of a 6 cm tall object 15 cm from a concave mirror of focal length 20 cm?

   Answer

   v = −60 cm (behind the mirror), m = +4: virtual, upright, 24 cm high

7. The magnification of a mirror is −0.333. Its image is formed on a card 540 mm from the mirror. Where is the object, and what is the focal length?

   Answer

   u = 1.62 m; f = +405 mm (concave)

23.2 Refraction at spherical surfaces

Refraction at a plane surface (revision)

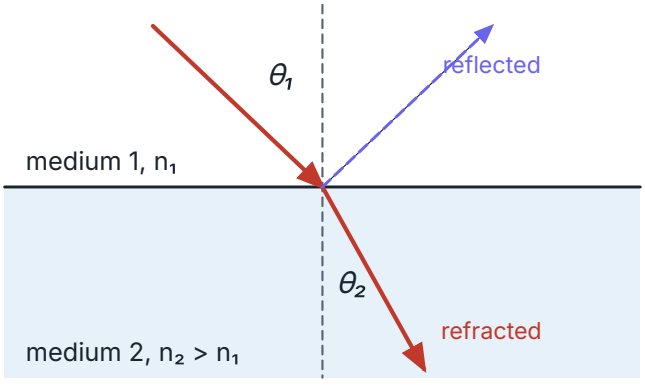


Figure 23.3: light going into a denser medium bends towards the normal.

- The incident ray, the refracted ray and the normal lie in the same plane.
- **Snell's law**: $\frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2}$ = constant.

- **Refractive index**: $n = \frac{c}{v}$ (speed in a vacuum / speed in the medium); $n \geq 1$, no unit.
- The frequency does not change across the boundary, but the speed and the wavelength do: $n_1\lambda_1 = n_2\lambda_2$, and $\lambda_n = \lambda_0/n$.

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

Example 23.2.1

Light of wavelength 500 nm in air strikes a slab of transparent material at 40.0° to the normal. The refracted beam makes 26.0° with the normal. Find the refractive index and the wavelength in the material.

Solution

$$n_2 = \frac{(1)\sin 40.0°}{\sin 26.0°} = \mathbf{1.47}$$

$$\lambda_2 = \frac{n_1\lambda_1}{n_2} = \frac{500}{1.47} = \mathbf{340\ nm}$$

Example 23.2.2

Light travelling in air strikes crown glass ($n = 1.52$) at 30.0° to the normal. Find the angle of refraction.

Solution

$$\sin\theta_2 = \frac{\sin 30.0°}{1.52} = 0.329 \Rightarrow \theta_2 = \mathbf{19.2°}$$

Example 23.2.3

A layer of oil (n = 1.45) floats on water (n = 1.33). Light strikes the oil at 40.0° to the normal. Find the angle the ray makes with the normal in the water.

Solution

$n_{air} \sin 40.0° = n_{oil} \sin \theta_{oil} = n_{water} \sin \theta_{water}$ (parallel boundaries)

$\sin \theta_{water} = \sin 40.0°/1.33 \Rightarrow \theta_{water} =$ **28.9°** (the oil does not affect the final angle)

Refraction at a spherical surface

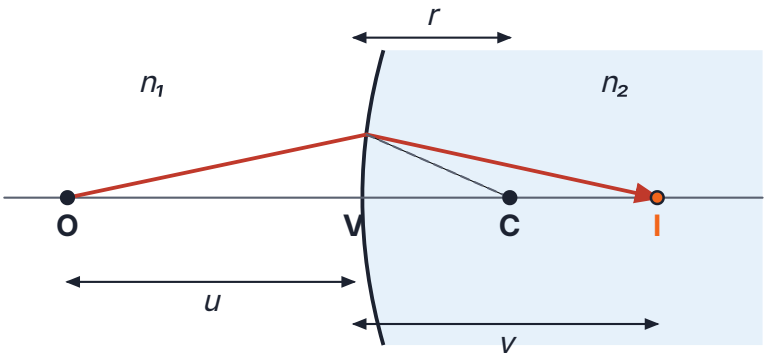


Figure 23.4: a point object O in medium n_1 forms an image I in medium n_2 by refraction at a spherical surface with centre C.

$$\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{r} \qquad m = \frac{h'}{h} = -\frac{n_1 v}{n_2 u}$$

n_1 = medium the light comes from; n_2 = medium the light goes into. (The derivation is not required.)

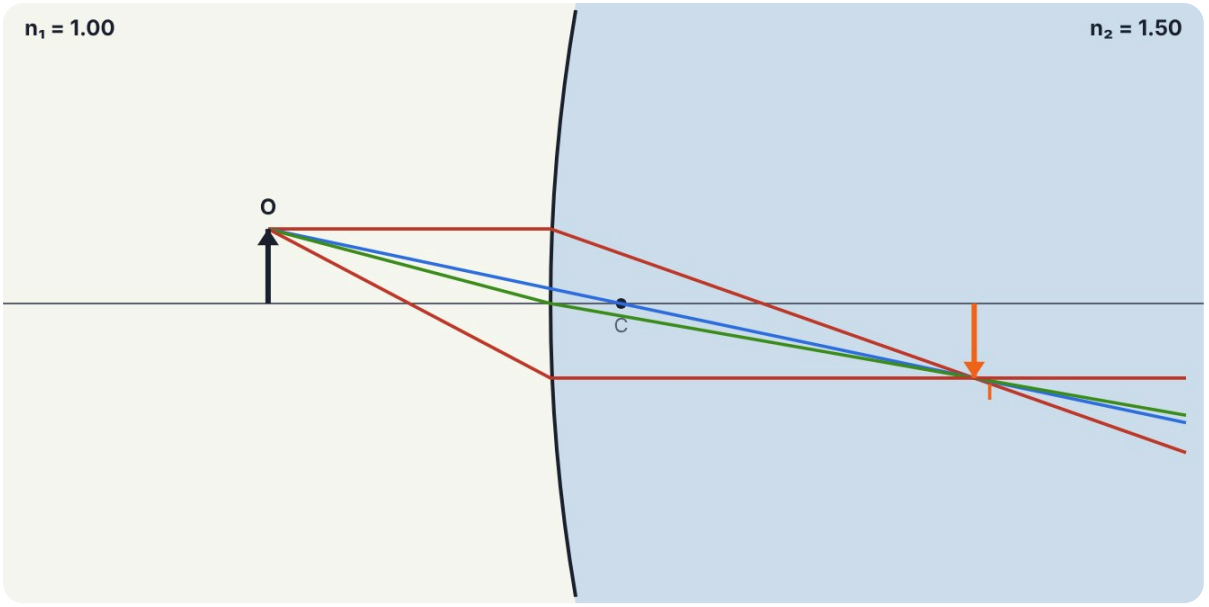
Quantity	Positive (+)	Negative (−)
u	real object (in front of the surface)	virtual object (behind the surface)
v	real image (behind the surface, in medium 2)	virtual image (in front of the surface, in medium 1)
r	centre of curvature behind the surface (surface convex to the incoming light)	centre of curvature in front of the surface (surface concave to the incoming light)
flat surface	$r \to \infty$, so the right-hand side is 0	

Some books use $|n_2 - n_1|/r$ with r positive when the centre of curvature is in the denser medium. Both forms are accepted in STPM. State the convention you use and never mix the two rules.

Simulation

Refraction at a spherical surface

Change the refractive indices, the radius of the surface and the object distance. The ray towards the centre of curvature crosses the surface without bending; the parallel ray is refracted. Try a fish in water seen from the air (flat surface).



$n_1/u + n_2/v = (n_2 - n_1)/r$	image distance v	magnification m	image
r = 10 cm	**60.0 cm**	**−1.00**	**real, inverted, same size**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 23 to use the simulation.

Example 23.2.4

A glass rod (n = 1.50) has one end ground to a hemispherical surface of radius 20 mm. A point object is on the axis, 80 mm to the left of the vertex, in air. Find (a) the image distance, (b) the magnification.

Solution

n_1 = 1, n_2 = 1.50, r = +20 mm (centre inside the glass), u = +80 mm

(a) $\frac{1}{80} + \frac{1.50}{v} = \frac{1.50 - 1}{20} \Rightarrow v =$ **+120 mm** (real, inside the rod)

(b) $m = -\frac{(1)(120)}{(1.50)(80)} =$ **−1.0**: same size, inverted

Example 23.2.5

The rod in Example 23.2.4 is placed in water (n = 1.33). Find the image distance and the magnification.

Solution

$\frac{1.33}{80} + \frac{1.50}{v} = \frac{1.50 - 1.33}{20} = 0.0085 \Rightarrow v =$ **−185 mm** (virtual, 185 mm in front of the vertex, in the water)

$m = -\frac{(1.33)(-185)}{(1.50)(80)} =$ **+2.05**: upright, magnified

Example 23.2.6

A coin is embedded in a spherical plastic paperweight of radius 3.0 cm (n = 1.50). The coin is 2.0 cm from the surface nearest the viewer. Find the position of its image.

Solution

Light goes from plastic to air: n_1 = 1.50, n_2 = 1.00, u = 2.0 cm. The centre of curvature is on the coin's side (in front of the surface), so r = −3.0 cm.

$\frac{1.50}{2.0} + \frac{1.00}{v} = \frac{1.00 - 1.50}{-3.0} = 0.1667 \Rightarrow v =$ **−1.7 cm**

Virtual image 1.7 cm inside the plastic: the coin looks nearer the surface than it is.

Example 23.2.7

A fish is 8.0 cm below the flat surface of water (n = 1.33). Find the apparent depth of the fish seen from directly above, and the magnification.

Solution

Flat surface: $r \to \infty$, so $\frac{1.33}{8.0} + \frac{1.00}{v} = 0 \Rightarrow v =$ **−6.0 cm** (virtual; apparent depth 6.0 cm)

$m = -\frac{(1.33)(-6.0)}{(1.00)(8.0)} =$ **+1.0**: upright, same size

In general, apparent depth = real depth/n.

STPM 2015 (repeat) · Paper 3

An object 5.0 mm high is 35.0 cm from the vertex of the spherical surface of a glass rod (n = 1.5). The image forms inside the rod, 50.0 cm from the vertex. (i) State the characteristics of the image. (ii) Find the radius of curvature. (iii) Where must the object be placed to give a virtual image 10.0 cm from the vertex?

Solution

(i) $m = -\frac{n_1 v}{n_2 u} = -\frac{50.0}{1.5 \times 35.0} = -0.952$: **real, inverted**, almost the same size ($h' \approx 4.8$ mm)

(ii) $\frac{1}{35.0} + \frac{1.5}{50.0} = \frac{0.5}{r} \Rightarrow 0.05857 = 0.5/r \Rightarrow r =$ **+8.54 cm**

(iii) v = −10.0 cm: $\frac{1}{u} - \frac{1.5}{10.0} = \frac{0.5}{8.54} \Rightarrow \frac{1}{u} = 0.2086 \Rightarrow u =$ **4.80 cm**

Exercise: refraction at spherical surfaces

1. A point object is 25.0 cm from the centre of a glass sphere of radius 5.0 cm (n = 1.50). Find the position of the image formed by (a) the first surface, (b) the second surface.

   Answer

   (a) u_1 = 20.0 cm, r = +5.0 cm: $\frac{1}{20.0} + \frac{1.50}{v_1} = \frac{0.50}{5.0} \Rightarrow v_1$ = +30.0 cm (20.0 cm beyond the sphere) (b) this is a virtual object for the back surface, u_2 = 10.0 − 30.0 = −20.0 cm, r = −5.0 cm: $\frac{1.50}{-20.0} + \frac{1}{v_2} = \frac{1 - 1.50}{-5.0} \Rightarrow v_2$ = +5.71 cm: a real image 5.71 cm from the back surface

23.3 Thin lenses

A lens is a transparent object with two refracting surfaces whose principal axes coincide. A **thin lens** is one whose thickness is small compared with its focal length.

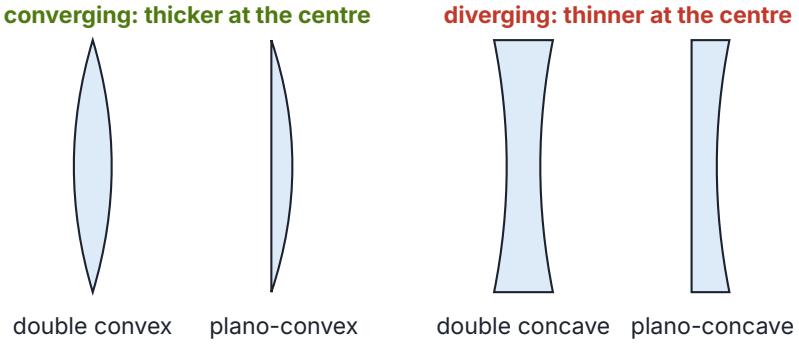


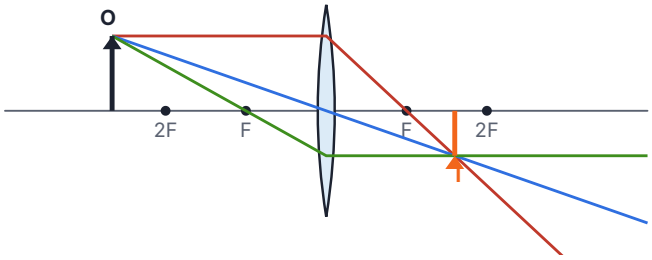
Figure 23.5: converging (convex) lenses make parallel rays converge; diverging (concave) lenses make them diverge.

- **Principal axis**: the line through the centre of the lens, perpendicular to both surfaces.
- **Focal point F**: the point where parallel paraxial rays meet after a converging lens, or from which they appear to diverge after a diverging lens. A lens has a focal point on each side.
- **Focal length f**: the distance between the focal point and the centre of the lens.

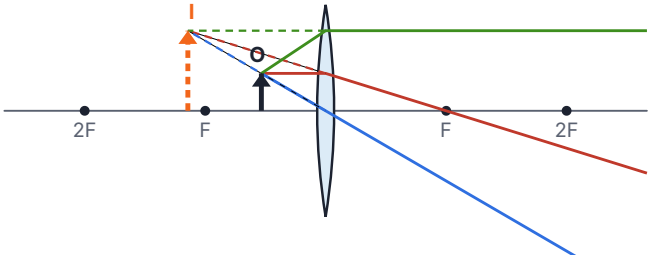
Ray diagrams for lenses

- **Ray 1**: a ray parallel to the axis passes through F on the other side (converging), or appears to come from F on the same side (diverging).
- **Ray 2**: a ray through F (or heading towards F on the far side, diverging) leaves parallel to the axis.
- **Ray 3**: a ray through the centre of the lens goes straight on.

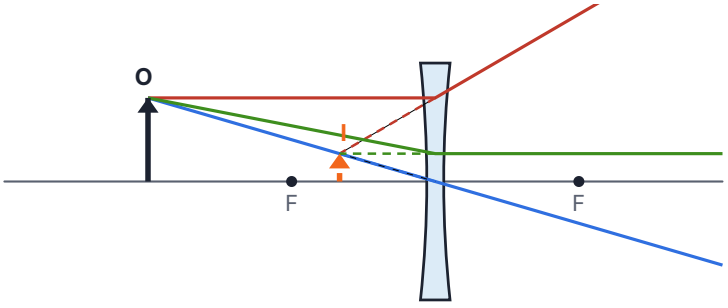
Object position (converging lens)	Image	Everyday use
beyond 2F	between F and 2F; real, inverted, diminished	camera, phone camera, the eye
at 2F	at 2F; real, inverted, same size	photocopier at 1 : 1
between F and 2F	beyond 2F; real, inverted, magnified	projector, enlarger
at F	at infinity	spotlight, searchlight
between F and the lens	on the same side; virtual, upright, magnified	magnifying glass



Converging lens, object beyond 2F.



Converging lens, object inside F: magnifying glass.



Diverging lens: the image is always virtual, upright and diminished, between the lens and F (door viewer, spectacles for short sight).

Thin lens formula and lensmaker's equation

Apply $\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{r}$ to each surface of a thin lens of refractive index n_l in a medium of index n_m.

Surface 1 (medium → lens): $\frac{n_m}{u} + \frac{n_l}{v_1} = \frac{n_l - n_m}{r_1}$ ... (1)

The image of surface 1 is the object for surface 2. For a thin lens it is a distance v_1 beyond surface 2, i.e. a virtual object: $u_2 = -v_1$.

Surface 2 (lens → medium): $\frac{n_l}{-v_1} + \frac{n_m}{v} = \frac{n_m - n_l}{r_2}$ … (2)

(1) + (2): $n_m\left(\frac{1}{u} + \frac{1}{v}\right) = (n_l - n_m)\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$

For an object at infinity, $v = f$. So the right-hand side divided by n_m is $1/f$, giving both results:

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f} \qquad m = -\frac{v}{u}$$

thin lens formula

$$\frac{1}{f} = \left(\frac{n_l}{n_m} - 1\right)\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

lensmaker's equation. In air, $n_m = 1$: $1/f = (n - 1)(1/r_1 - 1/r_2)$

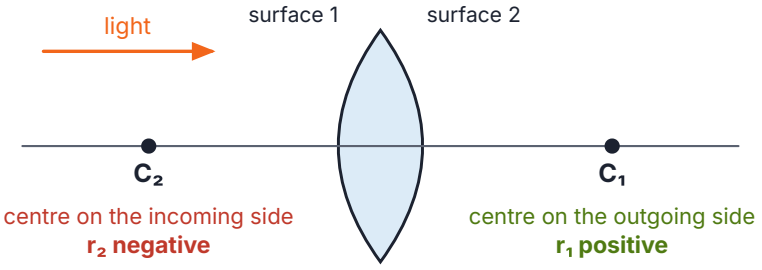


Figure 23.6: radius signs for a biconvex lens. r is positive when the centre of curvature is on the outgoing (back) side of that surface.

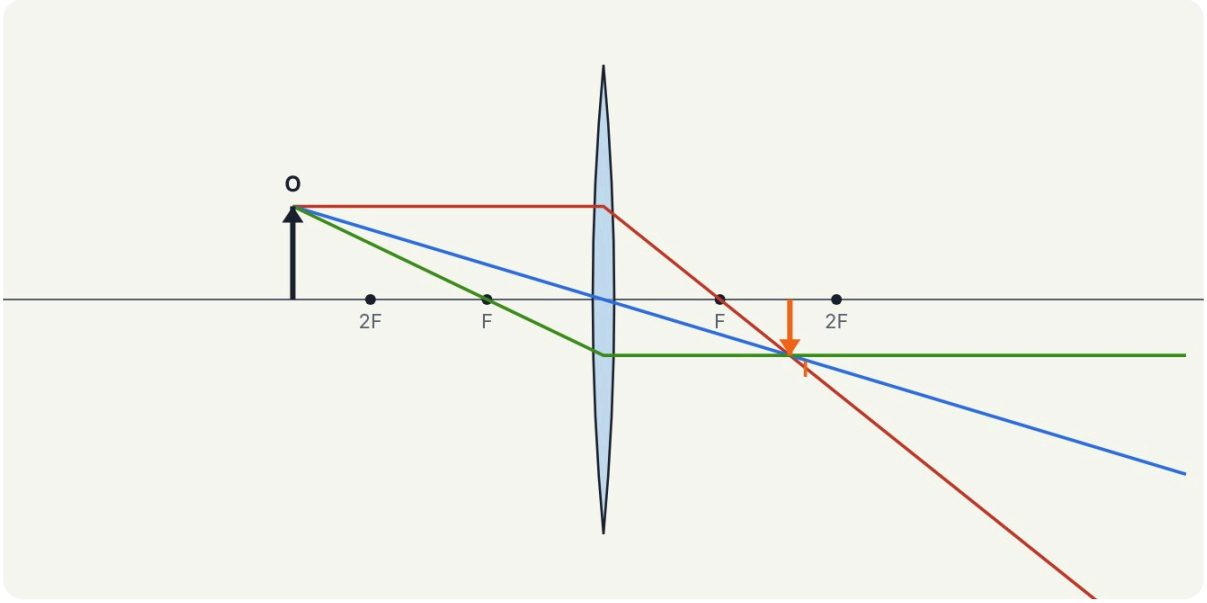
Quantity	Positive (+)	Negative (−)
f	converging lens	diverging lens
u	real object (in front of the lens)	virtual object (behind the lens)
v	real image (behind the lens)	virtual image (in front of the lens)
r	centre of curvature on the outgoing side	centre of curvature on the incoming side
m	upright image	inverted image

Biconvex: $r_1 = +$, $r_2 = -$. Biconcave: $r_1 = -$, $r_2 = +$. Flat face: $r = \infty$. Some books write $(1/r_1 + 1/r_2)$ with both radii as magnitudes for a biconvex lens; it is the same physics with a different convention. These notes use signed radii and the minus form throughout.

Simulation

Ray diagram for a thin lens

Choose a converging or diverging lens and move the object. The three principal rays find the image. Put the object at 2F, between F and 2F, and inside F, and compare with the table.



f	image distance v	magnification m	image
15.0 cm	**24.0 cm**	**−0.60**	**real, inverted, diminished**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 23 to use the simulation.

Example 23.3.1

An object is 10 cm from a converging lens of focal length 15 cm. Find the image position and its characteristics.

Solution

$\frac{1}{v} = \frac{1}{15} - \frac{1}{10} \Rightarrow v =$ **−30 cm** (virtual, on the same side as the object)

$m = -(-30)/10 =$ **+3.0**: upright, three times as large (a magnifying glass)

Example 23.3.2

Where must a small insect be placed if a diverging lens of focal length 25 cm is to form a virtual image 20 cm in front of the lens?

Solution

$f = -25$ cm, $v = -20$ cm: $\frac{1}{u} = \frac{1}{-25} - \frac{1}{-20} = \frac{1}{100} \Rightarrow u =$ **100 cm**

Example 23.3.3

A diverging meniscus lens has a focal length of −16 cm. The lens is held 10 cm from an object. Find the image position and the magnification.

Solution

$\frac{1}{v} = \frac{1}{-16} - \frac{1}{10} \Rightarrow v =$ **−6.15 cm**

$m = 6.15/10 =$ **+0.615** (virtual, upright, diminished)

Example 23.3.4

An object 450 mm from a converging lens forms a real image 900 mm from the lens. Find the focal length.

Solution

$\frac{1}{f} = \frac{1}{450} + \frac{1}{900} = \frac{3}{900} \Rightarrow f = 300$ mm = **30 cm**

Example 23.3.5

An object 6 cm high is 4 cm from a diverging meniscus lens of focal length −24 cm. Find the nature, size and location of the image.

Solution

$$\frac{1}{v} = \frac{1}{-24} - \frac{1}{4} \Rightarrow v = \mathbf{-3.43\ cm}$$

m = 3.43/4 = 0.857 ⇒ h' = 0.857 × 6 = **5.14 cm**; virtual and upright

Exercises: thin lens formula

1. An object 8 cm high is 30 cm from a converging lens of focal length 12 cm. Find the nature, size and location of the image.

   Answer

   v = +20 cm, m = −0.667: real, inverted, 5.33 cm high (h' = −5.33 cm)

2. A 1.70 m tall person stands 2.50 m in front of a camera whose converging lens has a focal length of 0.0500 m. Find the characteristics of the image on the sensor.

   Answer

   v = 0.0510 m, m = −0.0204: real, inverted, diminished, 0.0347 m high

3. An object is 18 cm in front of a diverging lens of focal length −12 cm. How far in front of the lens should the object be placed so that the size of its image is halved?

   Answer

   At 18 cm, m = 0.40; for m = 0.20: u = 48 cm

4. How far from a 50.0 mm focal length lens must an object be placed for a real image magnified 3.00×? And for a virtual image magnified 3.00×?

   Answer

   66.7 mm (real); 33.3 mm (virtual)

5. A lens focuses an object 22.5 cm away as an image 33.0 cm on the other side. What type of lens is it, what is its focal length, and is the image real or virtual?

   Answer

   converging, f = 13.4 cm, real

6. (a) A 2.70 cm high insect is 2.20 m from a lens of focal length 135 mm. Where is the image, how high is it, and what type is it? (b) Repeat for $f = -135$ mm.

Answer

(a) 144 mm behind the lens, real, inverted, 1.77 mm (b) 127 mm in front of the lens, virtual, upright, 1.56 mm

Using the lensmaker's equation

Example 23.3.6

A meniscus lens (n = 1.52) in air has radii r_1 = +22.4 cm and r_2 = +46.2 cm. Find its focal length and state the type of lens.

Solution

$\frac{1}{f} = (1.52 - 1)(\frac{1}{22.4} - \frac{1}{46.2}) = 0.52 \times 0.02300 \Rightarrow f =$ **+83.6 cm**

f is positive: a **converging** lens.

Example 23.3.7

Each face of a double convex lens has radius 20.0 cm; n = 1.50. Find the focal length (a) in air, (b) in carbon disulfide (n = 1.63).

Solution

r_1 = +20.0 cm, r_2 = −20.0 cm

(a) $\frac{1}{f} = (1.50 - 1)(\frac{1}{20.0} + \frac{1}{20.0}) \Rightarrow f =$ **+20.0 cm**

(b) $\frac{1}{f} = (\frac{1.50}{1.63} - 1)(0.100) = -0.00798 \Rightarrow f =$ **−125 cm**: in a denser liquid, the lens is **diverging**.

Example 23.3.8

A plano-convex lens of glass (n = 1.5) is to have a focal length of 40 cm. Find the radius of the curved surface.

Solution

Flat face first: $r_1 = \infty$. $\frac{1}{40} = (1.5 - 1)(0 - \frac{1}{r_2}) \Rightarrow r_2 = -20$ cm

radius of the curved surface = **20 cm**

STPM 2015 (repeat) · Paper 3

A thin biconvex lens (n = 1.5, both radii 10.0 cm) is immersed in a liquid of refractive index 1.65. Find the focal length and state how the lens behaves.

Solution

$\frac{1}{f} = (\frac{1.5}{1.65} - 1)(\frac{1}{10.0} + \frac{1}{10.0}) = (-0.0909)(0.200) \Rightarrow f =$ **−55 cm**

f is negative: in this liquid the lens **diverges** a parallel beam.

Power of a lens

Power of a lens

the reciprocal of its focal length in metres. Unit: dioptre (D) = m^{-1}. Converging lenses have positive power; diverging lenses have negative power.

$$P = \frac{1}{f}$$

Two thin lenses in contact

Lens 1: $\frac{1}{u} + \frac{1}{v_1} = \frac{1}{f_1}$. Its image is a virtual object for lens 2 (no gap): $u_2 = -v_1$.

Lens 2: $-\frac{1}{v_1} + \frac{1}{v} = \frac{1}{f_2}$. Adding: $\frac{1}{u} + \frac{1}{v} = \frac{1}{f_1} + \frac{1}{f_2}$

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} \qquad P = P_1 + P_2$$

Exercises: power and lenses in contact

1. A converging lens has a focal length of 12.5 cm. Calculate its power.

   Answer

   +8.0 D

2. A lens has a power of −4.0 D. State the type of lens and find its focal length.

   Answer

   diverging, f = −25 cm

3. A lens of power +6.0 D is in contact with a lens of power −2.0 D. Find the power and focal length of the combination.

   Answer

   +4.0 D, f = +25 cm

4. Two thin lenses of focal lengths +20.0 cm and +30.0 cm are in contact. Find the focal length and power of the combination.

   Answer

   f = +12.0 cm, P = +8.33 D

5. A double convex lens (n = 1.50) has r_1 = +15.0 cm and r_2 = −15.0 cm. Find its focal length and power in air.

   Answer

   f = +15.0 cm, P = +6.67 D

Combination of two lenses (separated)

Find the image I_1 formed by the first lens alone.

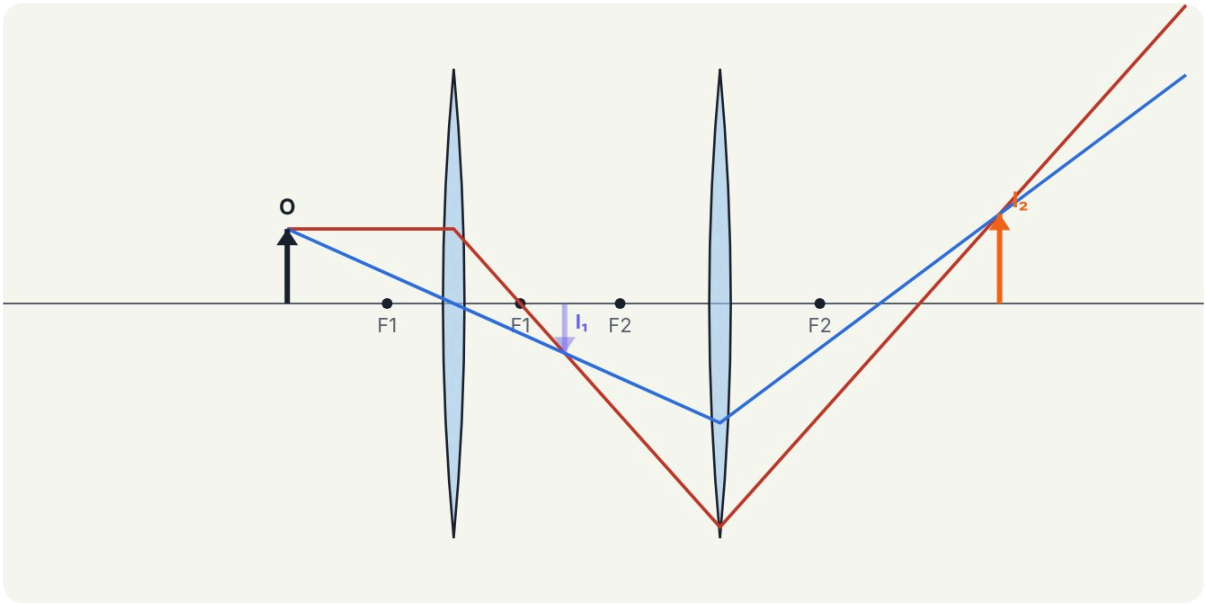
Treat I_1 as the object for the second lens: $u_2 = d - v_1$. If I_1 is in front of lens 2, u_2 is positive (real object). If I_1 falls beyond lens 2, u_2 is negative (virtual object).

Find the final image from lens 2. The overall magnification is $m = m_1 \times m_2$.

Simulation

Two-lens system

Set the two focal lengths, the separation and the object distance. The intermediate image I_1 (faint) becomes the object for the second lens. Make I_1 fall beyond lens 2 to see a virtual object.



lens 1: v_1	lens 2: $u_2 = d - v_1$	lens 2: v_2	$m = m_1 \times m_2$
16.7 cm	**23.3 cm**	**42.0 cm**	**1.20**

image
real, upright, magnified

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 23 to use the simulation.

Example 23.3.9

Two converging lenses, f_1 = 10.0 cm and f_2 = 20.0 cm, are 20.0 cm apart. An object is 30.0 cm in front of the first. Find the position and magnification of the final image.

Solution

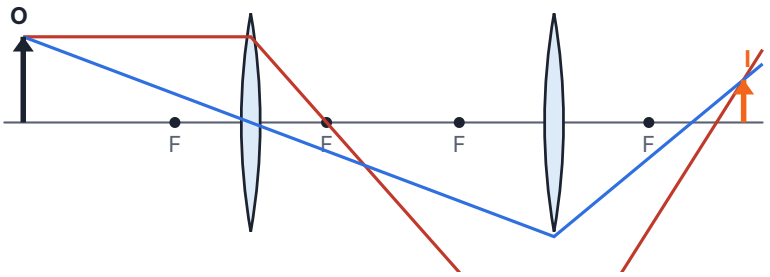
Lens 1: $\frac{1}{v_1} = \frac{1}{10.0} - \frac{1}{30.0} \Rightarrow v_1 = 15.0$ cm, $m_1 = -0.500$

Lens 2: $u_2 = 20.0 - 15.0 = 5.0$ cm; $\frac{1}{v_2} = \frac{1}{20.0} - \frac{1}{5.0} \Rightarrow v_2 =$ **−6.67 cm** (virtual, 6.67 cm in front of lens 2)

$m_2 = -(-6.67)/5.0 = +1.33$; $m = (-0.500)(+1.33) =$ **−0.667** (inverted)

Example 23.3.10

Two converging lenses, f_1 = 20.0 cm and f_2 = 25.0 cm, are 80.0 cm apart. An object is 60.0 cm in front of the first. Find the position and magnification of the final image.



Solution

Lens 1: $v_1 = 30.0$ cm, $m_1 = -0.500$

Lens 2: $u_2 = 80.0 - 30.0 = 50.0$ cm $\Rightarrow v_2 =$ **+50.0 cm** beyond lens 2 (real); $m_2 = -1.00$

$m = (-0.500)(-1.00) =$ **+0.500** (upright relative to the object)

Exercises: two lenses

1. A converging lens has focal length 0.080 m. An object is 0.040 m to its left. A second identical lens is 0.120 m to the right of the first. Where is the final image, relative to the second lens?

   Answer

   v_1 = −8.0 cm (virtual), u_2 = 20.0 cm ⇒ v_2 = +13.3 cm = 0.133 m to the right of lens 2; m = (+2.00)(−0.667) = −1.33

2. A coin is 20.0 cm to the left of a converging lens (f = 16.0 cm). A second identical lens is placed to the right so that the final image has the same size and orientation as the coin. Find the separation of the lenses.

   Answer

   v_1 = 80.0 cm, m_1 = −4.00; need m_2 = −0.250 ⇒ u_2 = 80.0 cm; separation = 160 cm

3. Two converging lenses 60 cm apart have focal lengths 10 cm and 15.0 cm. An object 6.0 cm high is 20 cm in front of the first. Find the location and size of the final image. Is it real or virtual?

   Answer

   v_1 = 20 cm, u_2 = 40 cm ⇒ v_2 = 24 cm beyond lens 2; m = (−1.00)(−0.600) = +0.600, h' = 3.6 cm; real

Combination of a lens and a spherical mirror

- Light passes through the lens, reflects from the mirror, then passes through the lens **again**. So there are three stages (lens → mirror → lens), and the three magnifications multiply.
- The image formed by the lens is the object for the mirror. If that image would fall behind the mirror, the mirror intercepts the rays before they meet: it is a **virtual object** and its object distance is negative.

Example 23.3.11

A converging lens (f = 10.0 cm) and a concave mirror (r = 30.0 cm) are 50.0 cm apart on a common axis. An object is 30.0 cm in front of the lens. Find the position and magnification of the final image.

Solution

Lens: $\frac{1}{v_1} = \frac{1}{10.0} - \frac{1}{30.0} \Rightarrow v_1 = 15.0$ cm, $m_1 = -0.500$

Mirror (f = +15.0 cm): $u_2 = 50.0 - 15.0 = 35.0$ cm $\Rightarrow v_2 = 26.25$ cm, $m_2 = -0.750$

Lens again: $u_3 = 50.0 - 26.25 = 23.75$ cm $\Rightarrow v_3$ = **17.3 cm** from the lens (on the object side, real); $m_3 = -0.727$

$m = (-0.500)(-0.750)(-0.727)$ = **−0.273** (inverted, diminished)

Exercises: lens and mirror

1. A converging lens (f = 15.0 cm) is 40.0 cm from a convex mirror of radius 20.0 cm on a common axis. An object is 30.0 cm in front of the lens. Find the position and magnification of the final image.

   Answer

   v_1 = 30.0 cm, m_1 = −1.00; mirror u_2 = 10.0 cm, f = −10.0 cm ⇒ v_2 = −5.0 cm, m_2 = +0.500; lens again u_3 = 45.0 cm ⇒ v_3 = 22.5 cm, m_3 = −0.500; final image 22.5 cm from the lens, m = +0.25 (upright)

2. A converging lens (f = 10.0 cm) and a concave mirror (f = 20.0 cm) are 80.0 cm apart. An object is 15.0 cm in front of the lens. Find the position and magnification of the final image.

   Answer

   v_1 = 30.0 cm (m_1 = −2.00); u_2 = 50.0 cm ⇒ v_2 = 33.3 cm (m_2 = −0.667); u_3 = 46.7 cm ⇒ v_3 = 12.7 cm (m_3 = −0.272); final image 12.7 cm from the lens, m = −0.36 (inverted)

STPM past-year questions

STPM 2013 · Paper 3 · Q7

A baby is in front of a concave mirror of radius of curvature R. An upright image twice the size of the baby is seen. What is the distance between the baby and the mirror?

A. $R/4$

B. $R/2$

C. $3R/4$

D. R

Answer

Answer: A

Upright ⇒ virtual, $m = +2 \Rightarrow v = -2u$. $\frac{1}{u} - \frac{1}{2u} = \frac{2}{R} \Rightarrow u = R/4$.

STPM 2013 · Paper 3 · Q8

A diverging lens and a converging lens, each of focal length 10.0 cm, are 10.0 cm apart, with the diverging lens in front. The arrangement is focused on a distant object. The image formed is

A. virtual, 5.0 cm behind the converging lens

B. virtual, 10.0 cm behind the diverging lens

C. real, 10.0 cm behind the converging lens

D. real, 20.0 cm behind the converging lens

Answer

Answer: D

Parallel rays give a virtual image at the focus of the diverging lens, 10.0 cm in front of it. So $u_2 = 20.0$ cm for the converging lens: $\frac{1}{20.0} + \frac{1}{v} = \frac{1}{10.0} \Rightarrow v = +20.0$ cm (real).

STPM 2014 · Paper 3 · Q7

A parallel beam is incident normally on a converging lens of focal length f_1 in contact with a diverging lens of focal length f_2. The focal length of the combination is f_3. If the image distances of the converging lens and of the diverging lens are v_1 and v_2, which pair (v_1, v_2) is correct?

A. (f_1, f_2)

B. (f_1, f_3)

C. (f_2, f_1)

D. (f_3, f_1)

Answer

Answer: B

The converging lens alone would focus the parallel beam at its own focus, so $v_1 = f_1$. Together the lenses act as one lens of focal length f_3, so the light converges at $v_2 = f_3$.

STPM 2015 · Paper 3 · Q7

An object is 5.0 cm in front of a concave mirror of radius of curvature 20.0 cm. Which statement about the image is true?

A. It is enlarged, upright and 10 cm behind the mirror.

B. It is enlarged, inverted and 10 cm in front of the mirror.

C. It is diminished, upright and 6.7 cm behind the mirror.

D. It is diminished, inverted and 6.7 cm in front of the mirror.

Answer

Answer: A

$f = +10$ cm: $\frac{1}{5.0} + \frac{1}{v} = \frac{1}{10} \Rightarrow v = -10$ cm; $m = +2$ (virtual, upright, enlarged).

STPM 2015 (repeat) · Paper 3 · Q7

A converging lens forms an inverted image twice the size of the object when the object is 30 cm from the lens. What is the new object distance if the image is also twice the size of the object but upright?

A. 5.0 cm

B. 10.0 cm

C. 15.0 cm

D. 20.0 cm

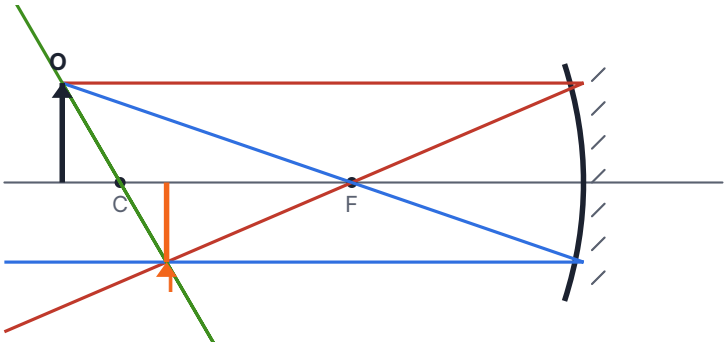
Answer

Answer: B

Inverted, $m = -2$ at $u = 30$ cm $\Rightarrow v = 60$ cm, $f = 20$ cm. Upright $\Rightarrow$ virtual, $v = -2u$: $\frac{1}{u} - \frac{1}{2u} = \frac{1}{20}$ $\Rightarrow u = 10.0$ cm.

STPM 2014 · Paper 3 · Q19(a)

A concave mirror has radius of curvature 40.0 cm. A nail 5.00 cm long stands upright on the principal axis, 45.0 cm in front of the mirror. (i) Calculate the image distance. (ii) Find the size of the image. (iii) State two characteristics of the image.



Solution

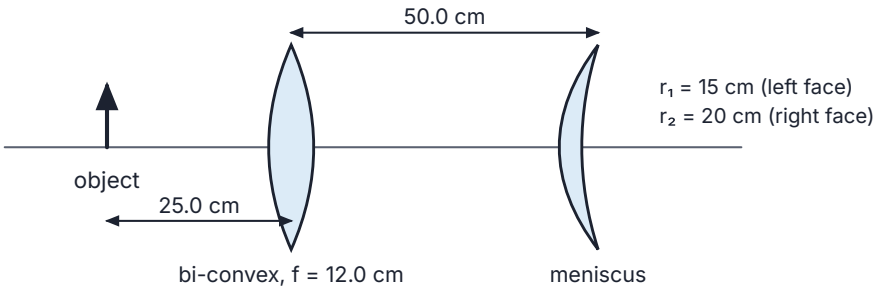
(i) $f = +20.0$ cm: $\frac{1}{v} = \frac{1}{20.0} - \frac{1}{45.0} \Rightarrow v =$ **+36.0 cm**

(ii) $m = -36.0/45.0 = -0.800 \Rightarrow$ image height = **4.00 cm**

(iii) real (v positive) and inverted (m negative); also diminished

STPM 2014 · Paper 3 · Q19(b)

An object is 25.0 cm in front of a biconvex lens of focal length 12.0 cm. A meniscus lens (radii 15.0 cm and 20.0 cm, n = 1.5) is 50.0 cm behind the biconvex lens. Determine (i) the focal length of the meniscus lens, (ii) the position of the final image, (iii) the magnification.



The arrangement in STPM 2014 Paper 3 Q19(b) (not to scale).

Solution

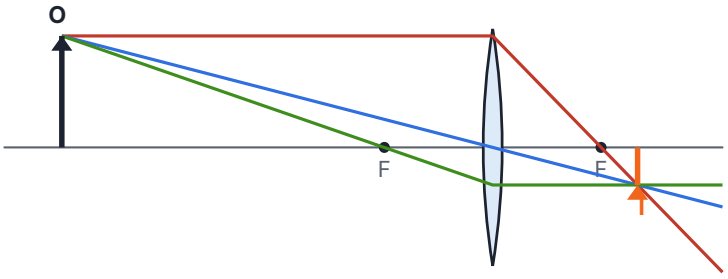
(i) From the figure, both faces of the meniscus bulge towards the incoming light, and the left face (radius 15.0 cm) is the more curved. Both centres of curvature are on the outgoing side, so r_1 = +15.0 cm, r_2 = +20.0 cm: $\frac{1}{f} = 0.5\left(\frac{1}{15.0} - \frac{1}{20.0}\right) \Rightarrow f =$ **+120 cm** (converging: thicker at the centre)

(ii) Biconvex lens: $\frac{1}{25.0} + \frac{1}{v_1} = \frac{1}{12.0} \Rightarrow v_1$ = +23.08 cm. Meniscus: u_2 = 50.0 − 23.08 = 26.92 cm $\Rightarrow \frac{1}{v_2} = \frac{1}{120} - \frac{1}{26.92} \Rightarrow v_2 =$ **−34.7 cm**: a virtual image 34.7 cm in front of the meniscus lens

(iii) $m = \left(-\frac{23.08}{25.0}\right)\left(-\frac{-34.71}{26.92}\right) =$ **−1.19** (inverted, slightly enlarged)

STPM 2015 · Paper 3 · Q19

(a) An object is 30.0 cm in front of a converging lens of focal length 7.55 cm. Find the image position, the magnification and the characteristics of the image. (b) A 10.0 cm object is 15.0 cm in front of a convex mirror of radius 40.0 cm. Find the image position and height. (c) A spider 2.00 cm across hangs 200 cm from a wall. A lens forms an image of the spider on the wall half its size. What lens is needed, and where?



Solution

(a) $\frac{1}{v} = \frac{1}{7.55} - \frac{1}{30.0} \Rightarrow v =$ **+10.1 cm**; $m = -10.1/30.0 =$ **−0.336**: real, inverted, diminished

(b) $f = -20.0$ cm: $\frac{1}{v} = \frac{1}{-20.0} - \frac{1}{15.0} \Rightarrow v =$ **−8.57 cm** (behind the mirror); $h' = 10.0 \times 8.57/15.0$ = **5.71 cm**: virtual, upright, diminished

(c) The image is on a screen, so it is real: a **converging** lens. $|m| = v/u = ½ \Rightarrow u = 2v$, and $u + v =$ 200 cm $\Rightarrow v = 66.7$ cm from the wall, $u = 133$ cm; $\frac{1}{f} = \frac{1}{133} + \frac{1}{66.7} \Rightarrow f =$ **44.4 cm**